Lecture 3: Operating Systems

Processes, threads and scheduling

Prof. Example - Winter 2026

What is a process?

A process is a program in execution.

Each process has its own address space.

The kernel keeps a process control block (PCB).

States: new, ready, running, waiting, terminated.

Threads

A thread is the unit of CPU scheduling.

Threads share code, data and open files.

Each thread has its own stack and registers.

Context switches between threads are cheaper.

CPU Scheduling

First-Come First-Served (FCFS)

Shortest Job First (SJF) minimises average waiting time.

Round Robin uses a fixed time quantum.

Priority scheduling can cause starvation; fix it with aging.